

Harshvardhan Mani Tripathi

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🌐 LinkedIn

🐙 GitHub

🔗 LeetCode

EDUCATION

Indian Institute of Information Technology, Surat
Bachelor of Technology in Computer Science And Engineering

2022 - 2026
CGPA - 7.68

EXPERIENCE

21CC Education

Software Development Engineer Intern

Jan 2026 – Present

Mumbai, Maharashtra

- Owned **end-to-end development** of **10+** backend-driven web pages for the **CRISP Client** in the **21CC Backoffice** using **Groovy on Grails**, delivering **Employer Onboarding**, **Mass Job/Location Creation**, **Candidate Discovery** via **Google Geocoding API**, and a **Resume Parsing** module using **Tesseract OCR** and **Gemini LLM** to automate structured data extraction.
- Driving migration of the **21CC App** from **Angular** to **React**; independently implemented **5+** production features including **Vimeo optimization** (load time: 4s → 1–1.5s), **Resume Upload/Parsing**, **Recruiter Inbox**, **Google libphonenumber** integration for **global phone validation**, **Leaderboard dashboards**, and **Job Vacancy map** via **Google Maps**.

Vysion Technology

Machine Learning Intern — Bio-medical Waste Object Detection

May 2025 – July 2025

Jodhpur, Rajasthan

- Built a **real-time bio-medical waste detection system** using **Python**, **YOLO**, and **OpenCV**, managing the **full ML pipeline** — data collection, annotation via **Roboflow**, augmentation, training, and **quantitative evaluation**.
- Achieved **70%+ mAP** through **systematic hyperparameter tuning**, **optimized model architecture design**, and rigorous testing across **diverse real-time detection scenarios**.

PROJECTS

Smart Face Recognition Attendance System | Python, Flask, React, OpenCV, Scikit-learn, SQLite, JWT, ESP32 [GitHub](#)

- Built a **full-stack attendance platform** with a **Flask REST API** and **React** frontend, enforcing **role-based access control** secured with **JWT** authentication and **Bcrypt** hashing.
- Developed a **real-time face recognition pipeline** using **OpenCV** and **KNN (Scikit-learn)**, integrating **ESP32 IoT** hardware triggers to automate attendance logging with **zero manual intervention** in an **SQLite** database.

DocTalk – Medical Chatbot | Python, Flask, MySQL, LangChain, HTML, CSS, JavaScript, GCP

[GitHub](#)

- Engineered and **deployed to production** an AI medical chatbot on **GCP** using **Flask** and **Docker** with a **Retrieval-Augmented Generation (RAG)** architecture for **context-aware, document-grounded** responses.
- Powered **semantic Q&A** via **LangChain** and **Google Gemini**, processing medical documents with **Sentence-Transformers** and storing vectors in **Pinecone** for **low-latency similarity search**.

Lofi Music Directory | React, Node.js, Express, MongoDB, Tailwind CSS, Howler.js, Cloudinary

[GitHub](#)

- Engineered a **full-stack music streaming platform** with **JWT** and **Bcrypt** auth, **uninterrupted cross-route playback** via **Howler.js** and **React Context API**, and **scalable dynamic audio/thumbnail delivery** via **Cloudinary** cloud storage.

TECHNICAL SKILLS

Programming Languages: Python, Groovy, C++, JavaScript, C, SQL, HTML, CSS

Libraries / Frameworks: Flask, FastAPI, Grails, Node.js, React, Next.js, Express, TensorFlow, PyTorch, LangChain, Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib

Tools & Platforms: Git, Docker, Google Cloud Platform, Render, PostgreSQL, VS Code, Jupyter Notebook, Google Colab, Claude, Gemini, Notion

Domains: Web Development, Machine Learning, Deep Learning, LLMs

Software Engineering: Agile, Scrum, Git Workflows, CI/CD Pipelines, API Development

CS Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, OOP

ACHIEVEMENTS AND LEADERSHIP

- Innovation Grant:** Secured an **INR 2.5 Lakh grant** under the **SSIP** scheme by **i-Hub**, Government of Gujarat.
- Secretary, Ruminant E-Cell & Head, E-Summit 2025, IIIT Surat:** Led a **40+** member team and organized **10+** events/hackathons/workshops with **1000+** attendees.
- DevHeat Beta Hackathon 2024:** Selected among the **Top 8 teams** out of **35+** competing teams in a development hackathon, demonstrating strong **full-stack development** and **rapid prototyping** skills.
- Smarth India Hackathon 2023:** Qualified for the **second round** with a project focused on developing a **Doctor Scheduling Platform**.
- Competitive Programming:** Solved **350+** problems across LeetCode and GeeksforGeeks.
- RoboSoccer, IIT Surat Techfest:** Won **1st place** among **20+** competing teams, demonstrating **robotics, teamwork**, and **competitive engineering** skills.

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